Worksheet: Propositional Logic (Topic 1, Part 1)

Introduction

Welcome to the first part of Topic 1, Logic. In this module we meet propositional logic, the system that deals with propositions (statements) and the connectives that combine them. You will learn to recognise a proposition, to translate everyday sentences into logical formulae and back again, to build truth tables, and to classify formulae as tautologies, consistent formulae or contradictions.

A motivating problem: on an island live knights, who always tell the truth, and liars, who always lie. You meet two people, A and B. A says, "At least one of us is a liar," and B says nothing. What can you conclude? You will have the tools to solve this riddle by the end of the topic.

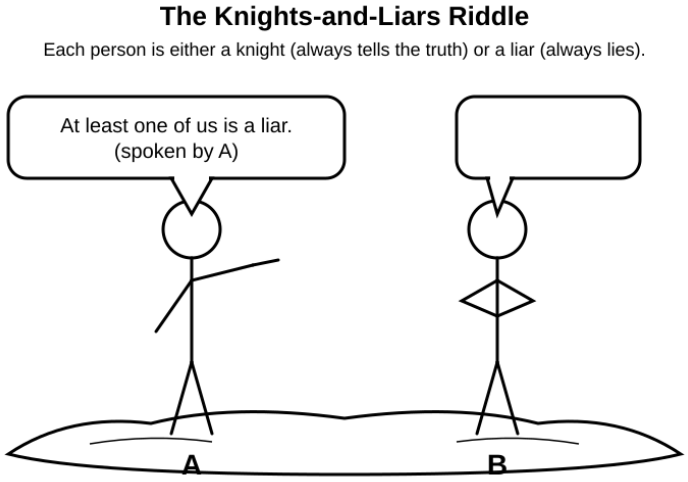


Figure 1

Figure 1. The knights-and-liars riddle that motivates this topic: A makes a claim, B stays silent.

By the end of this worksheet you will be able to:

- decide whether a sentence is a proposition and assign it a truth value;
- translate between English sentences and well-formed formulae using the six connectives;
- construct truth tables for compound formulae, respecting operator precedence;
- classify a formula as a tautology, a consistent formula or a contradiction, with and without full truth tables.

Estimated time: approximately 60 minutes.

How to use this worksheet: work through the activities in order, use the hints if you are stuck, and make a genuine attempt at each one before looking at the answer sheet. Activity 4 is the most involved; if you are running short, ensure you complete (a) and (b) before attempting (c).

Key terms and notation

Term / symbol	Meaning
Proposition (statement)	A declarative sentence that is either true or false, but not both and not neither.
Truth value	True, written T or 1, or false, written F or 0.
Atomic proposition	A specific statement labelled with a capital letter such as P, Q, R.
Propositional variable	A lowercase letter such as p, q, r standing for a statement whose truth value is not specified; used to carry on a general argument.
Well-formed formula (wff)	A string built from atomic propositions and the six connectives, with brackets placed so that every connective has exactly two operands (or one, for negation).
Negation	$\neg p$: true if and only if p is false.
Conjunction	$p \wedge q$: true only if both p and q are true.
Disjunction	$p \vee q$: true if at least one of p or q is true.
Implication (conditional)	$p \to q$: true if p is false or q is true; p is the premise, q the conclusion.
Biconditional	$p \leftrightarrow q$: true if p and q have the same truth value.
Exclusive or	$p \oplus q$: true if exactly one of p or q is true.
Truth table	A table listing every combination of truth values for the variables and the resulting value of a formula. For n variables it has 2^n rows.
Tautology	A formula that is always true.
Consistent formula	A formula that is not a contradiction; that is, it is true for at least one assignment of truth values.
Contradiction (inconsistent formula)	A formula that is never true.

Operator precedence, highest first: $\neg$, then $\wedge$, then $\vee$, then $\rightarrow$, and finally $\leftrightarrow$.

> **Common pitfall.** An implication $p \rightarrow q$ is false in only one case: p true and q false. Think of it as a promise, "If you study hard, I will give you a cookie." The promise is broken only if you do study hard and the cookie does not appear; in every other case the promise stands.

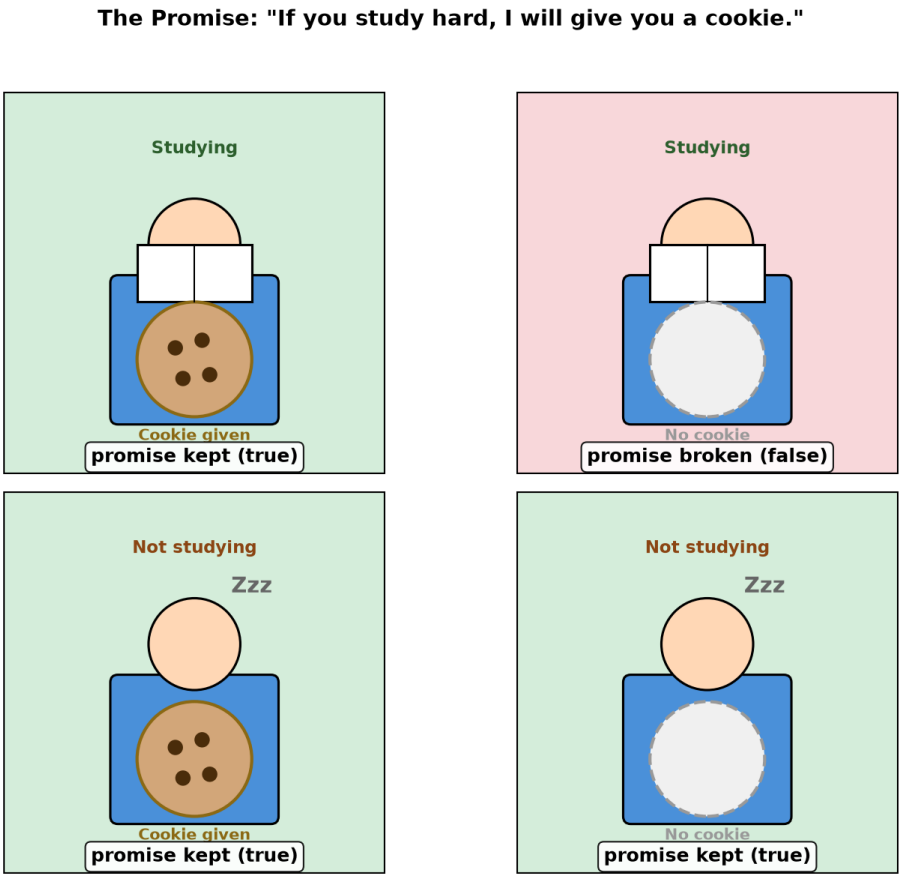


Figure 2

Figure 2. The promise analogy: the implication is false only when the premise holds but the conclusion does not.

> **Gotcha.** Keep atomic propositions free of connectives. In "the UK does not leave the EU", label P as "the UK leaves the EU" and write $\neg P$, rather than labelling the whole negative sentence as one proposition.

Note the distinction between the two kinds of letter: capital letters (P, Q) name specific statements, while lowercase letters (p, q) are general propositional variables.

Activities

Activity 1

Start point. A proposition is a declarative sentence with a definite truth value; questions, imperatives and sentences whose truth value depends on a variable are not propositions. Specific statements are labelled with capital letters; general propositional variables use lowercase letters.

Instructions.

(a) For each sentence below, write "proposition" or "not a proposition", and give a one-line reason.

1. 2 is a prime number.
2. 5 is an even number.
3. x is a prime number.
4. Are you going to school?
5. Do your homework now.

(b) Let P be "3 is an odd number" and Q be "$2 + 5$ is an even number". State the truth value of P and of Q.

Expected output. Five labelled classifications with reasons, and two truth values.

Answers: see Activity 1 in the answer sheet. Do not look before making a proper attempt.

Activity 2

Start point. Connectives combine atomic propositions into compound ones. You translate a formula into English by reading each connective, and you translate English into a formula by first labelling the atomic (connective-free) statements.

Instructions.

(a) Given P = "I study 20 hours a week", Q = "I attend all the lectures", R = "I will pass the exam", S = "I will be happy", translate the formula $(P \vee Q) \to (R \wedge S)$ into a single English sentence.

(b) Translate the sentence "If the UK does not leave the EU, then skilled nurses would not leave the NHS and research grants will remain intact" into a well-formed formula. First list your atomic propositions, then build the formula step by step.

Expected output. One English sentence for (a); a list of atomic propositions and one final formula for (b).

> **Hint.** In (b), remember that "not" is itself a connective, so keep it out of the atomic labels.

Answers: see Activity 2 in the answer sheet. Do not look before making a proper attempt.

Activity 3

Start point. Each connective has a fixed truth table, and when a formula omits brackets we apply precedence: $\neg$ first, then $\wedge$, then $\vee$, then $\rightarrow$, and finally $\leftrightarrow$.

Instructions.

(a) Complete the result column for each connective below. The columns for p and q are given in the order 1,1; 1,0; 0,1; 0,0.

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \oplus q$
1	1					
1	0					
0	1					
0	0					

When filling in the $p \rightarrow q$ column, the promise-analogy figure above may help: the promise is broken in exactly one row.

(b) Using precedence only (no brackets are given), show how the formula $p \rightarrow p \wedge \neg q \vee q$ is parsed, writing one bracketing step per line, ending with the fully bracketed formula.

Expected output. A completed five-column table and a short sequence of bracketing steps.

Answers: see Activity 3 in the answer sheet. Do not look before making a proper attempt.

Activity 4

Start point. A truth table for a compound formula has one row per combination of its variables (2^n rows for n variables). You fill in intermediate columns first, then the main connective, and finally classify the formula.

Instructions.

(a) Build the full truth table for $(p \wedge q) \rightarrow p$, including a column for $p \wedge q$. State whether the formula is a tautology, consistent or a contradiction.

(b) Build the full truth table for $p \wedge (q \rightarrow p)$, including a column for $q \rightarrow p$. Compare its final column with the one from (a) and explain in one sentence why the two formulae differ.

(c) Build the full truth table for $(\neg p \wedge q) \leftrightarrow (p \vee \neg q)$, showing the columns $\neg p$, $\neg p \wedge q$, $\neg q$ and $p \vee \neg q$. Classify the formula.

Expected output. Three completed truth tables, each with a classification, plus one comparison sentence.

> **Hint.** For (c), the biconditional is true only where its two sides agree; check whether that ever happens.

Answers: see Activity 4 in the answer sheet. Do not look before making a proper attempt.

Activity 5

Start point. You can often classify a formula without a full truth table by replacing tautologies such as $p \vee \neg p$ with True, contradictions such as $p \wedge \neg p$ with False, and then using the useful relations (for example, tautology $\vee$ consistent is a tautology; contradiction $\rightarrow$ tautology is a tautology).

Instructions. For each formula, simplify step by step without a full truth table and state whether it is a tautology, consistent or a contradiction.

(a) $(p \vee \neg p) \vee (p \wedge \neg p)$

(b) $(p \rightarrow p) \vee (q \wedge r)$

(c) $(q \wedge (p \vee \neg p)) \rightarrow q$

(d) $(q \wedge (p \wedge \neg p)) \rightarrow ((q \wedge \neg q) \rightarrow (\neg p \wedge p))$

Expected output. For each item, a short chain of substitutions ending in True, False or a consistent formula, plus the classification.

> **Hint.** In (d), work from the innermost contradictions outward, and remember that "False $\rightarrow$ False" is a tautology.

Answers: see Activity 5 in the answer sheet. Do not look before making a proper attempt.

Wrap-up

In this worksheet you have classified sentences as propositions, translated between English and formulae, built truth tables for the six connectives and for compound formulae, respected operator precedence, and classified formulae as tautologies, consistent formulae or contradictions — both by full tables and by shortcut substitution. These ideas underpin the course objective of understanding logical arguments: the form of an argument is captured by an implication from its premise to its conclusion, and that implication is a tautology exactly when the argument is valid. In the next part of Topic 1 you will meet equivalencies and first-order logic, and you will return to the knights-and-liars riddle with a fuller toolkit. Recall the island scene at the start of this worksheet — you now have the tools to determine that A must be a knight and B a liar.

Answer sheet

Activity 1 (answer)

(a)

1. Proposition — it is a declarative sentence, and it is true.
2. Proposition — declarative, and false.
3. Not a proposition — its truth value depends on the value of x (true for $x = 2$, false for $x = 6$).
4. Not a proposition — it is a question, so it has no truth value.
5. Not a proposition — it is an imperative (an order), so it has no truth value.

(b) P = "3 is an odd number" is true (T). Q = "2 + 5 is an even number" is false (F), because 2 + 5 = 7, which is odd.

Activity 2 (answer)

(a) "If I study 20 hours a week or I attend all the lectures, then I will pass the exam and I will be happy."

(b) Atomic propositions (no connectives inside):

- P: the UK leaves the EU.
- Q: skilled nurses will leave the NHS.
- S: research grants will remain intact.

Step by step:

- "the UK does not leave the EU" becomes $\neg P$.
- "skilled nurses would not leave the NHS" becomes $\neg Q$.
- "and research grants will remain intact" becomes $\neg Q \wedge S$.
- The whole "if ... then ..." becomes $\neg P \rightarrow (\neg Q \wedge S)$.

Final formula: $\neg P \rightarrow (\neg Q \wedge S)$.

Gotcha. Do not label P as "the UK does not leave the EU". Because "not" is a connective, the atomic proposition must be the positive statement, with the negation applied outside it.

Activity 3 (answer)

(a)

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \oplus q$
1	1	1	1	1	1	0
1	0	0	1	0	0	1
0	1	0	1	1	0	1
0	0	0	0	1	1	0

Notes: $p \wedge q$ behaves like multiplication ($p \times q$); $p \vee q$ like the maximum; $p \to q$ is false only on the row $p = 1, q = 0$; $p \leftrightarrow q$ is an equality check; $p \oplus q$ is the opposite of the biconditional.

(b) Parsing $p \to p \wedge \neg q \vee q$:

1. Negation first: $p \to p \wedge (\neg q) \vee q$.
2. Conjunction next: $p \to (p \wedge \neg q) \vee q$.
3. Disjunction next: $p \to ((p \wedge \neg q) \vee q)$.

The implication is already the outermost connective, so no further bracketing is needed.

Fully bracketed: $p \to ((p \wedge \neg q) \vee q)$.

Activity 4 (answer)

(a) $(p \wedge q) \to p$:

p	q	$p \wedge q$	$(p \wedge q) \to p$
1	1	1	1
1	0	0	1
0	1	0	1
0	0	0	1

The final column is all 1, so the formula is a **tautology**.

(b) $p \wedge (q \to p)$:

p	q	$q \to p$	$p \wedge (q \to p)$
1	1	1	1
1	0	1	1
0	1	0	0
0	0	1	0

The final column is 1, 1, 0, 0, so this formula is **consistent** (not a tautology). It differs from (a) because in (a) the implication is the main connective (true whenever its premise $p \wedge q$ is false), whereas in (b) the conjunction is the main connective, so the whole formula is false whenever p is false. This is why precedence matters.

(c) $(\neg p \wedge q) \leftrightarrow (p \vee \neg q)$:

p	q	$\neg p$	$\neg p \wedge q$	$\neg q$	$p \vee \neg q$	$(\neg p \wedge q) \leftrightarrow (p \vee \neg q)$
1	1	0	0	0	1	0
1	0	0	0	1	1	0
0	1	1	1	0	0	0
0	0	1	0	1	1	0

The final column is all 0, so the formula is **inconsistent**, that is, a **contradiction**.

Gotcha. It is easy to read the biconditional as "or". Remember it is an equality check: it is true only where the two sides match, and here they never do.

Activity 5 (answer)

(a) $(p \vee \neg p) \vee (p \wedge \neg p)$.

- $p \vee \neg p$ is a tautology, so replace with True.
- $p \wedge \neg p$ is a contradiction, so replace with False.
- The formula becomes True $\vee$ False, which is True.
- Classification: **tautology**.

(b) $(p \to p) \vee (q \wedge r)$.

- $p \to p$ is a tautology, so replace with True.
- $q \wedge r$ is consistent.
- The formula becomes True $\vee$ consistent, which is True (tautology $\vee$ consistent is a tautology).
- Classification: **tautology**.

(c) $(q \wedge (p \vee \neg p)) \to q$.

- $p \vee \neg p$ is a tautology, so replace with True.
- The formula becomes $(q \wedge \text{True}) \to q$.
- $q \wedge$ True simplifies to q, giving $q \to q$.
- $q \to q$ is a tautology.
- Classification: **tautology**.

(d) $(q \wedge (p \wedge \neg p)) \to ((q \wedge \neg q) \to (\neg p \wedge p))$.

- $\neg p \wedge p$ is a contradiction, so replace with False.
- $q \wedge \neg q$ is a contradiction, so replace with False.
- The inner implication becomes False $\to$ False, which is a tautology, so replace with True.
- $p \wedge \neg p$ is a contradiction, so replace with False; the premise becomes $q \wedge$ False, which is a contradiction, so replace with False.
- The formula becomes False $\to$ True, which is a tautology.
- Classification: **tautology**.

Gotcha. In (d), do not stop at "False $\rightarrow$ False" and call the whole formula a contradiction. The inner "False $\rightarrow$ False" is itself always true, so it becomes True, and the outer implication then has a false premise, making the whole formula a tautology.